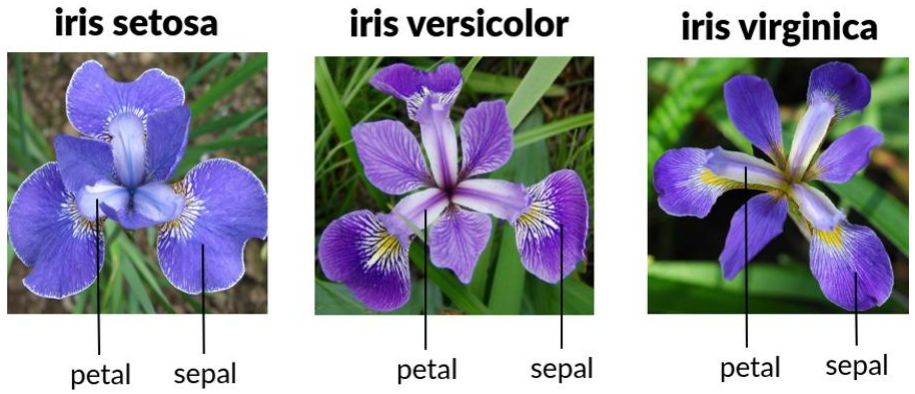
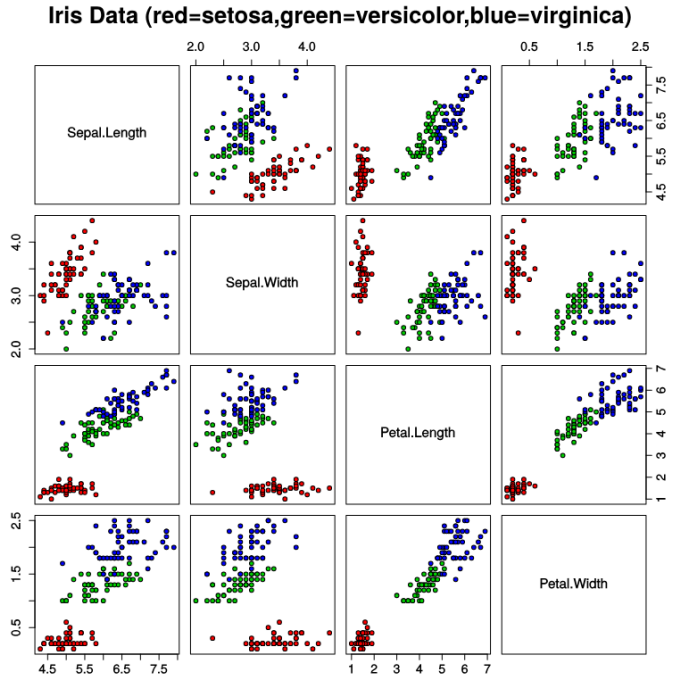
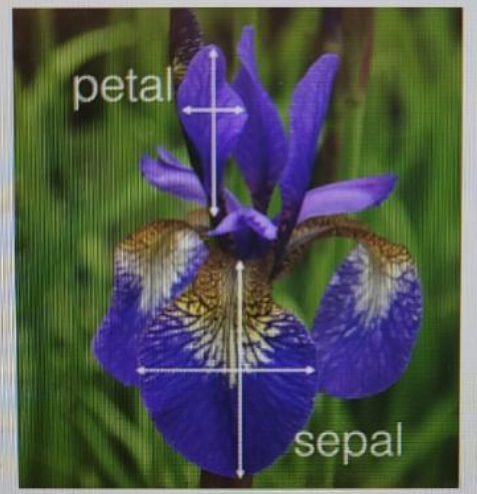
Iris dataset





4. In 1936 Ronald Fisher published a famous paper on classification titled "The use of multiple measurements in taxonomic problems." In the paper, Fisher study the problem of classifying iris flowers based on measurements of the sepal and petal widths and lengths, depicted in the image below.



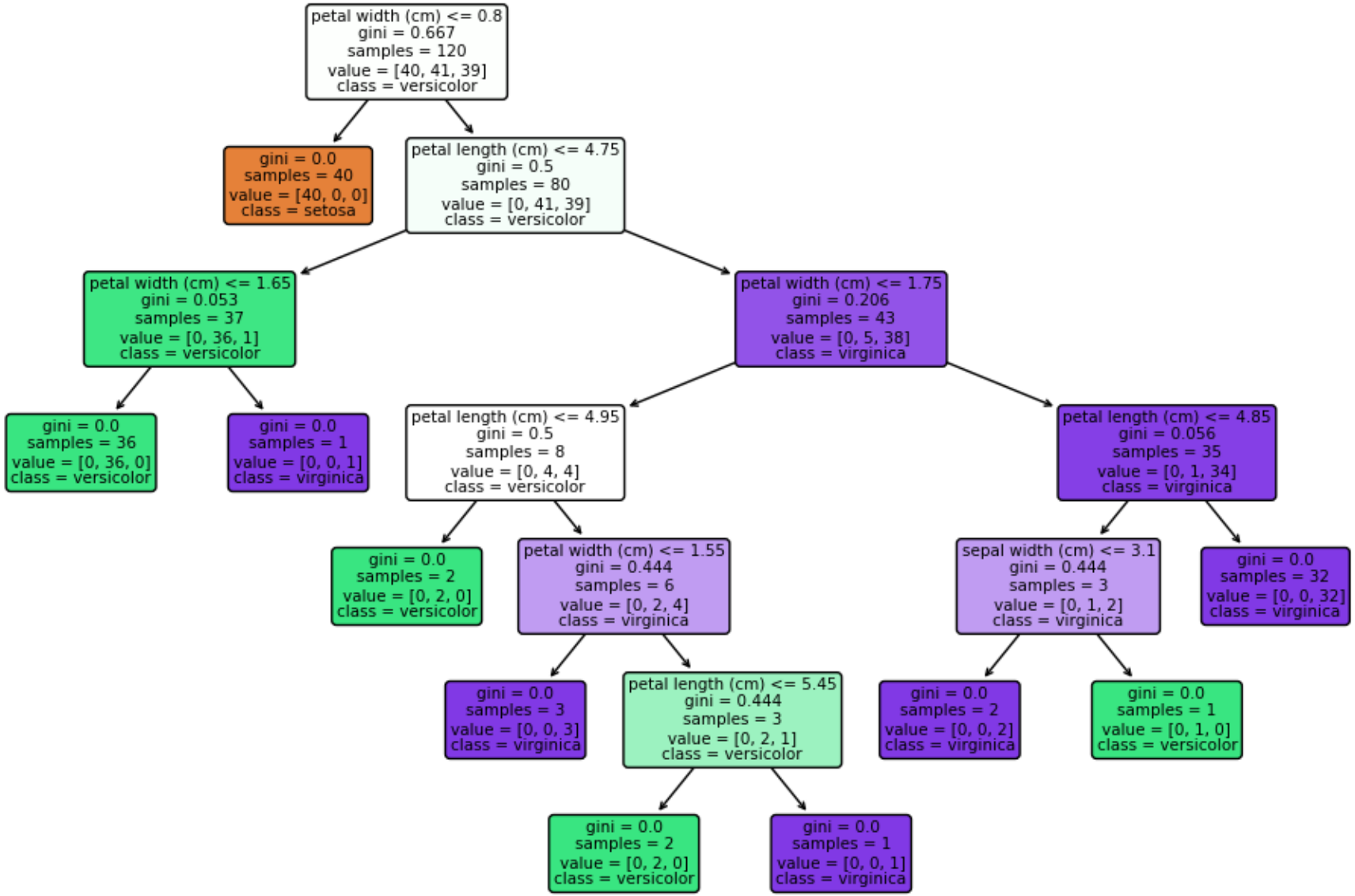
Fisher's dataset is available in Matlab (`fisheriris.mat`) and is widely available on the web (e.g., Wikipedia). The dataset consists of 50 examples of three types of iris flowers. The sepal and petal measurements can be used to classify the examples into the three types of flowers. Approach the iris classification problem using a *generative modeling* approach. Assume that each class-conditional density is MVN.

(a) Fit the MVN models to the training data using estimates of means and covariances. Design a classifier based on these fitted models.

(b) Constrain the MVN models so that the decision boundaries between pairs of classes are linear.

Fisher, R.A. (1936) The Use of Multiple Measurements in Taxonomic Problems. Annals Eugenics, 7, 179-188.

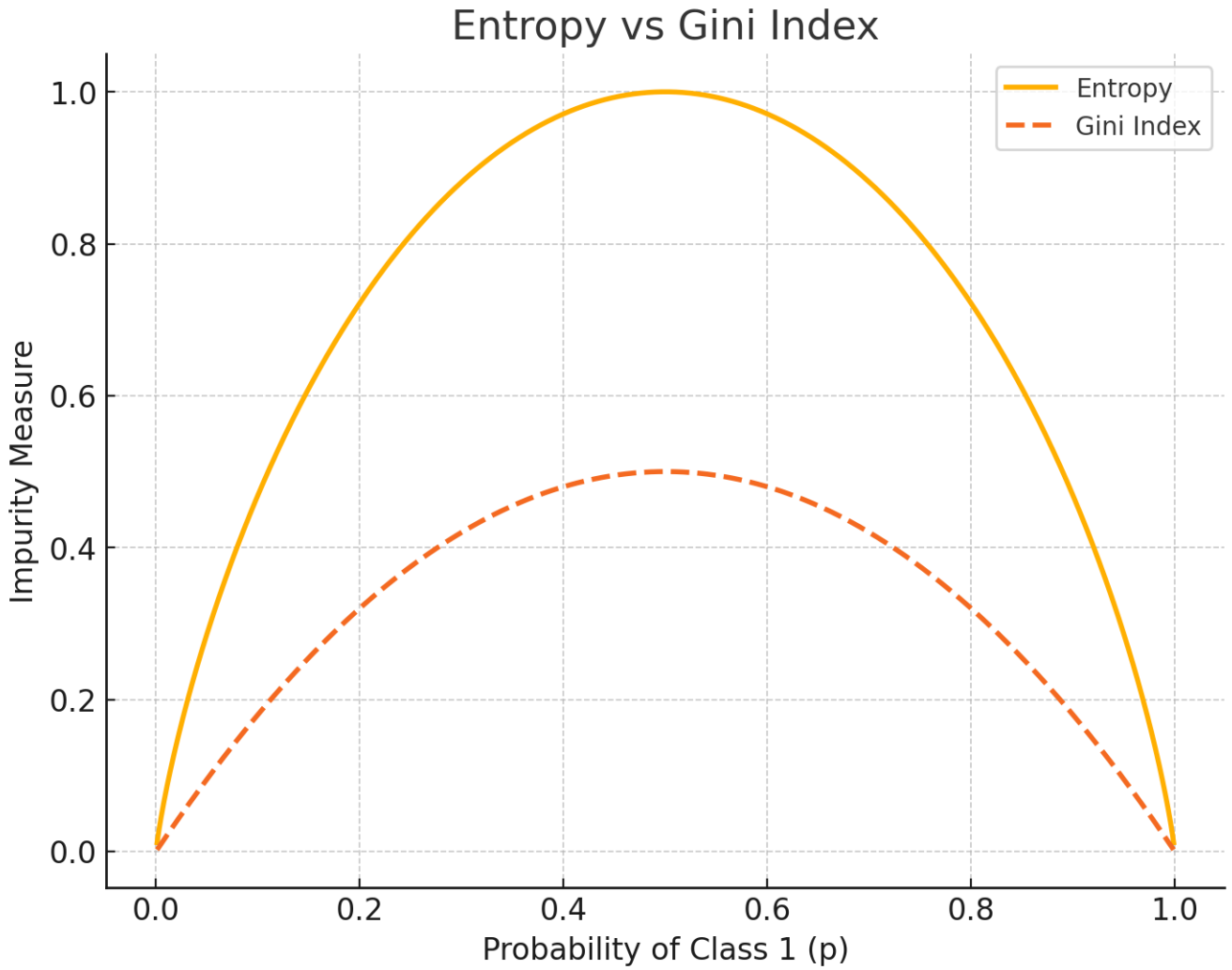
Decision tree



Decision tree formalization of the cost function

$$\text{Entropy} = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

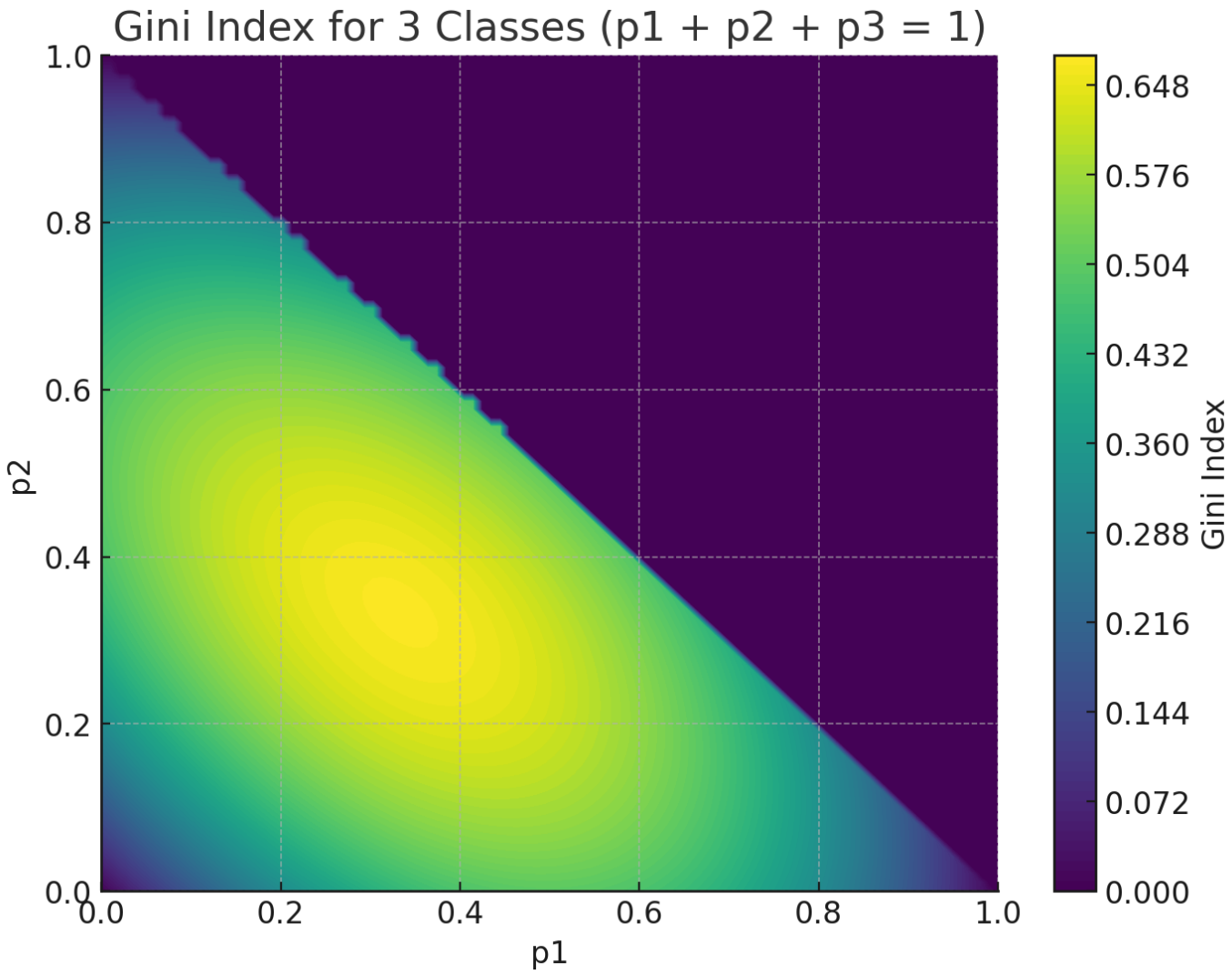
$$\text{Gini Index} = 1 - \sum_{i=1}^{C} p_i^2$$



Decision tree formalization of the cost function

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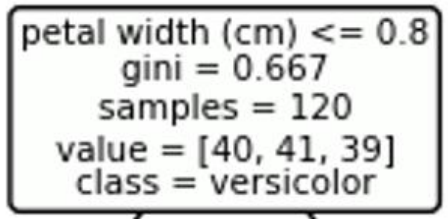
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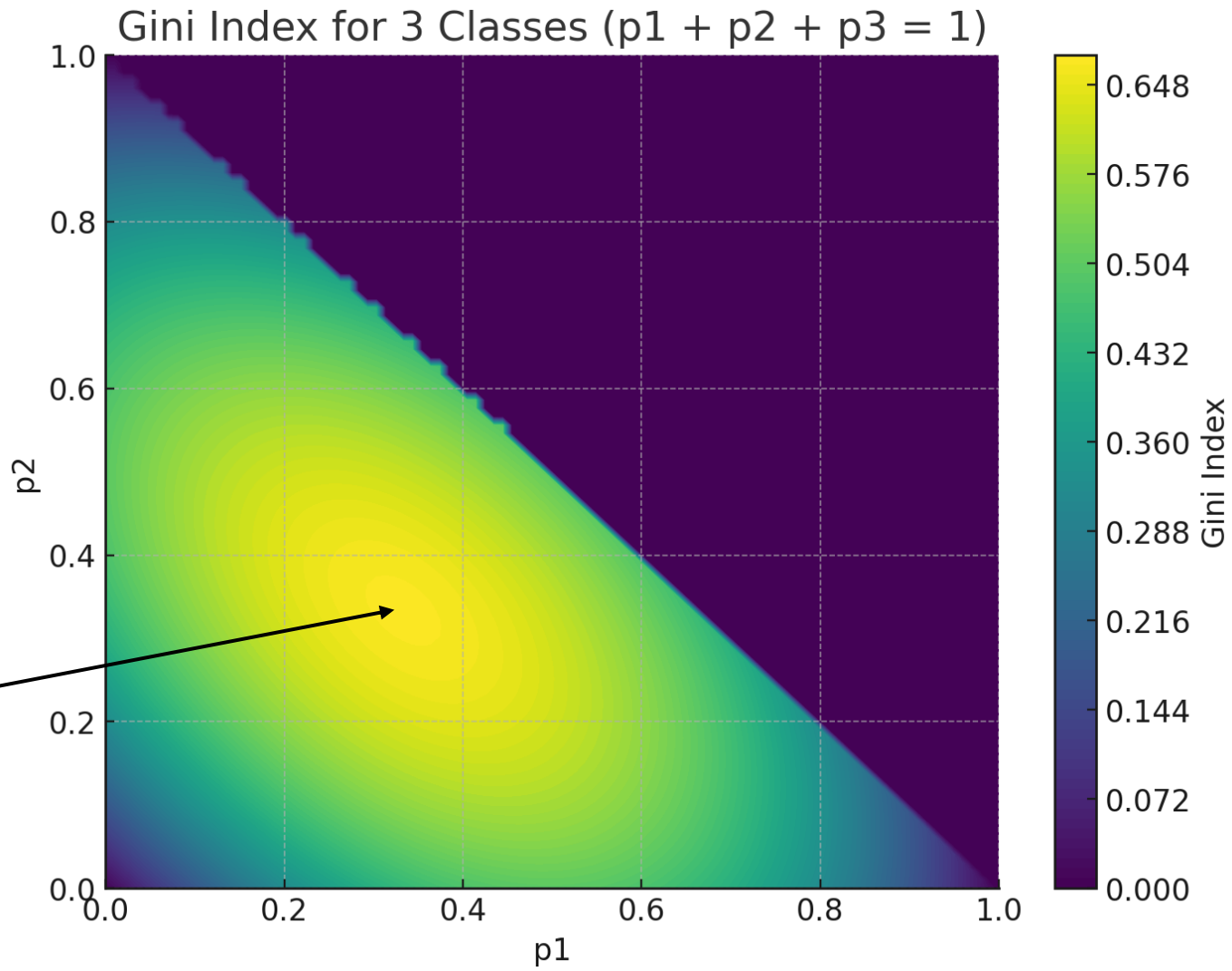


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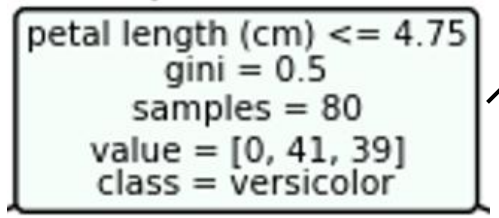


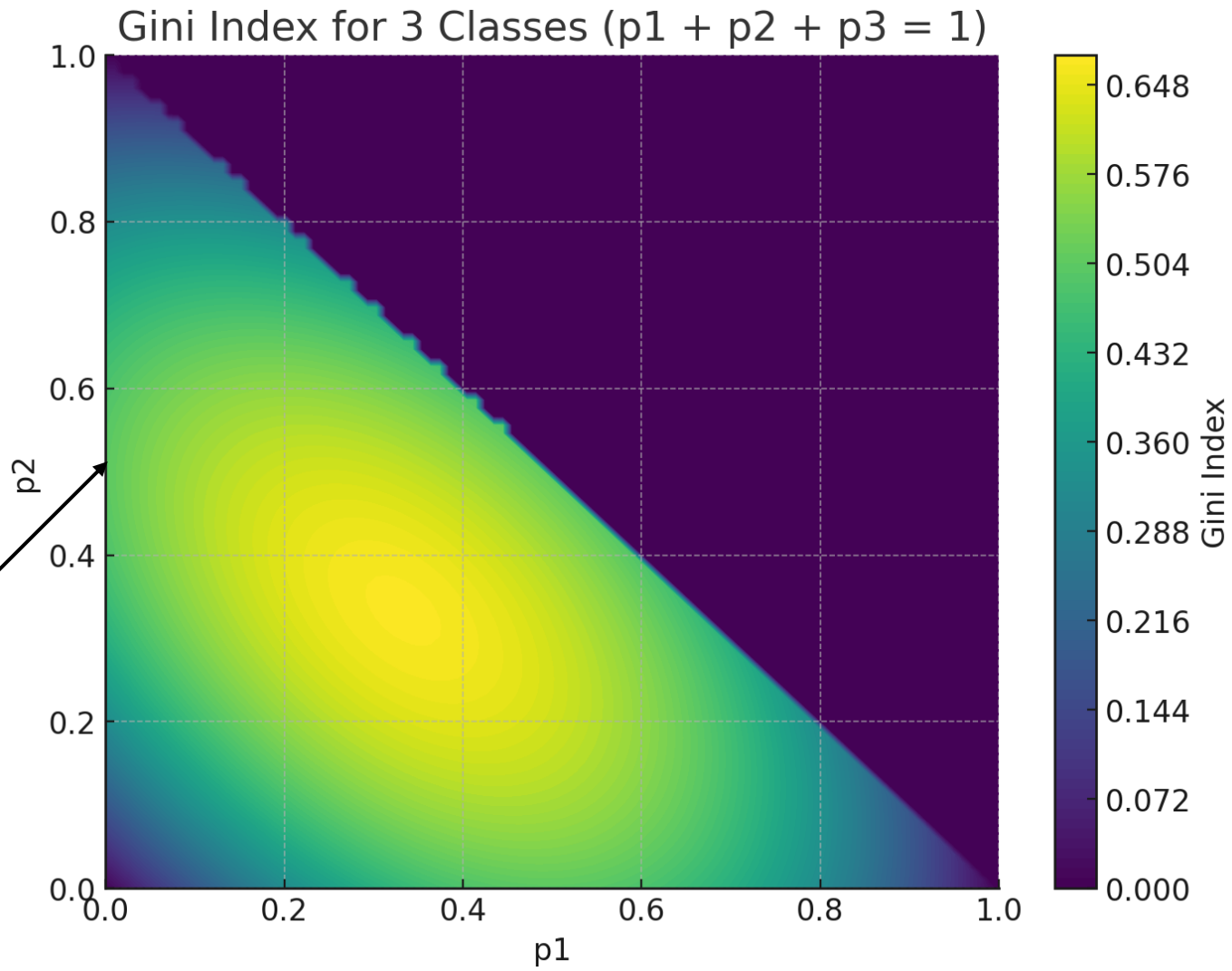


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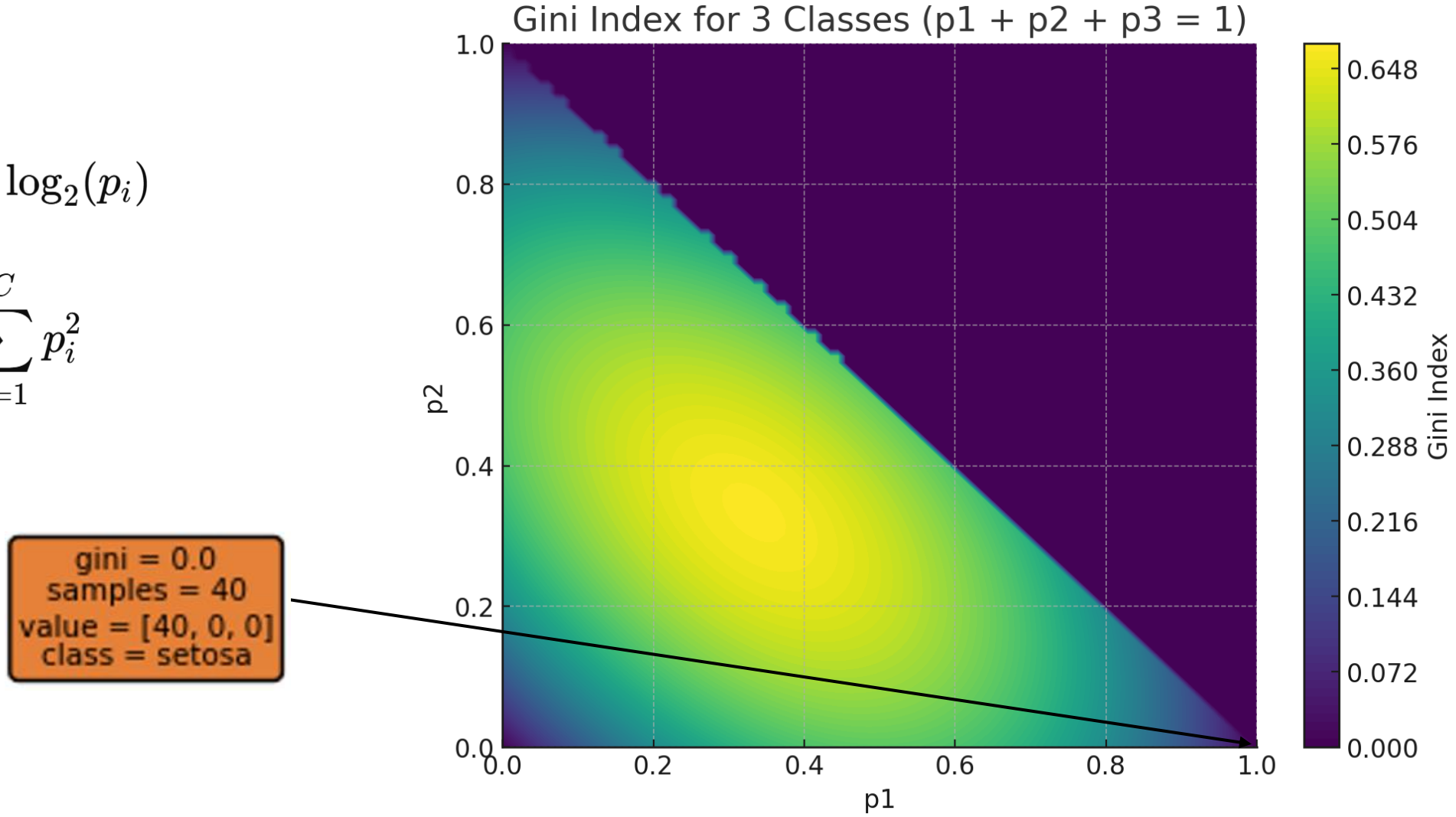




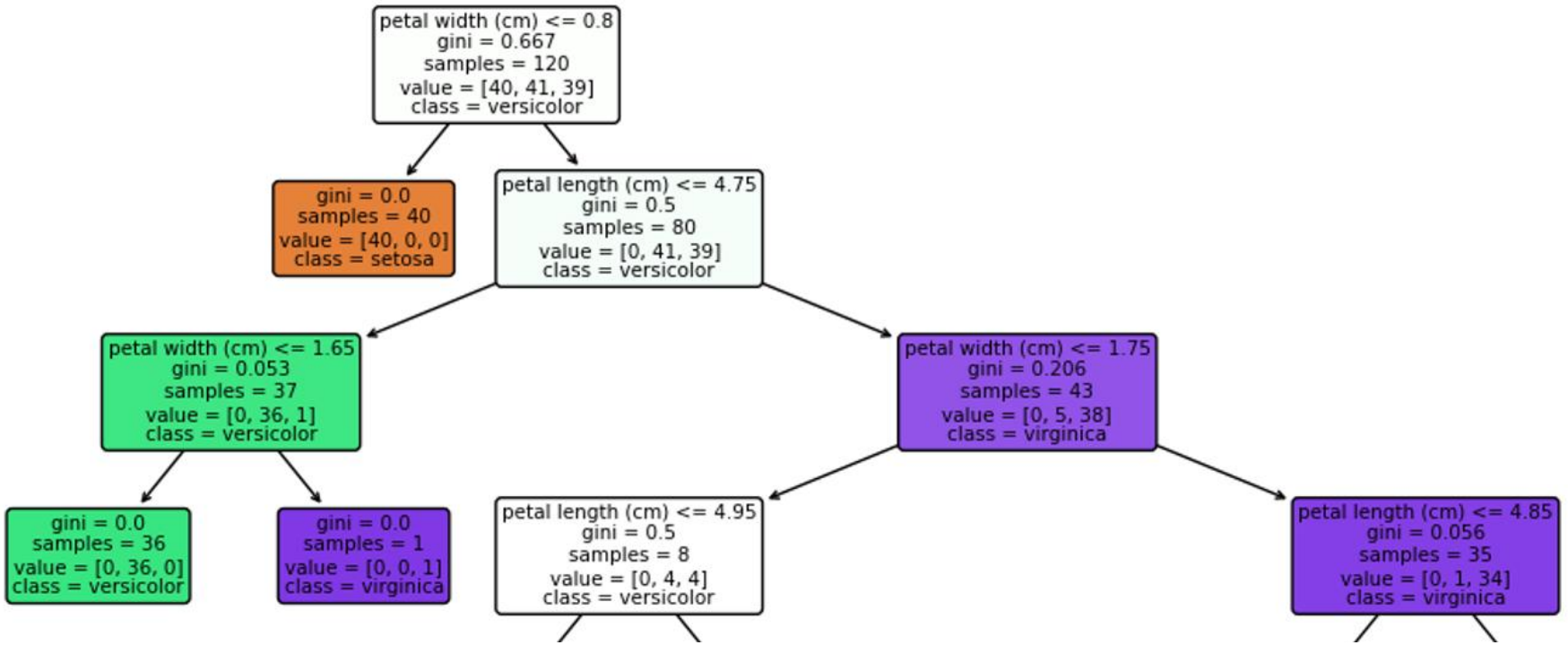
Decision tree formalization of the cost function

$$\text{Entropy} = -\sum_{i=1}^{C} p_i \log_2(p_i)$$

$$\text{Gini Index} = 1 - \sum_{i=1}^{C} p_i^2$$



Decision tree formalization of the cost function



For a dataset D split into two subsets D_{left} and D_{right}, the Gini criterion is:

$$\text{Gini}_{\text{split}} = \frac{|D_{\text{left}}|}{|D|} \cdot \text{Gini}(D_{\text{left}}) + \frac{|D_{\text{right}}|}{|D|} \cdot \text{Gini}(D_{\text{right}})$$